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🌐 [LinkedIn](#) | 🐙 [GitHub](#) | 📁 [Portfolio](#)

EDUCATION

Universitas Indonesia

Bachelor of Engineering in Computer Engineering; GPA: 3.75/4.00

Depok, Indonesia

Exp. Apr 2027

SMAN 34 Jakarta

Diploma of Mathematics and Sciences; Grade: 90/100

South Jakarta, Indonesia

July 2020 – Apr 2023

EXPERIENCE

National Taiwan University of Science and Technology (NTUST)

Research Intern, BMW Lab (TEEP Program)

Taipei, Taiwan / Remote

Jan 2026 – Present

- Completed remote O-RAN research interconnecting OSC SMO and FlexRIC over O1/A1/E2 interfaces for AI-driven QoS optimization, validating 29/29 integration tests.
- Built OAI PRACH simulator and RFSim baselines, mapped gNB detection thresholds, and developed a context-aware mitigation pipeline for log extraction, policy replay, and sensitivity testing with 44 passing tests.
- Reproduced controlled PRACH attacker runs on an on-site OAI UE/gNB testbed with USRP B210 hardware, resolving launch dependencies and preserving repeatable configuration, execution, and cleanup evidence.

PT. Pharos Indonesia

AI, Mechatronics & Full-Stack Software Intern

South Jakarta, Indonesia

Jan 2026 – Jun 2026

- Developed an Autoreject Blister inspection application using PyQt, OpenCV, YOLO, and Arduino integration, with interactive ROI/lane editing, CLAHE preprocessing, per-lane counters, product master data, and audit trail.
- Trained and compared YOLO models for pharmaceutical blister quality control, then integrated camera, proximity sensor, servo/rejector, and hardware-debug workflows for local validation.
- Built QR-BPOM production-scan features across a Next.js frontend and Go backend, including WebSocket device events, temporary Master Box flow, duplicate/unreadable QR handling, scan ID reset, and monitoring fixes.

Fakultas Teknik Universitas Indonesia (FTUI)

Laboratory & Teaching Assistant

Depok, Indonesia

Feb 2025 – Jan 2026

- Laboratory Assistant for Physics Lab:** Facilitated experiments twice a week and graded technical reports for mechanical and electrical physics modules.
- Teaching Assistant for Computational Thinking & Electrical Circuits:** Mentored and graded students in algorithmic logic and Python programming, and supported instruction in circuit analysis.

Ikatan Mahasiswa Elektro FTUI

Vice Head of Academics and Professions

Depok, Indonesia

Jan 2024 – Dec 2025

- Led strategic academic initiatives and professional development programs for the student body.
- Previously served as Staff of Academics (2024), coordinating peer tutoring and study resources.

PROJECTS

O.A.S.I.S. Water Purification System | *ESP32, IoT, Power Electronics, RO Filtration*

- Designed a portable, off-grid water purification system for post-disaster use, integrating RO filtration, ESP32-based pH/turbidity sensing, adaptive sanitation/drinking-water routing, and offline local monitoring.
- Developed the system architecture and feasibility analysis for a 12V VRLA power train, buck/boost conversion, pumps, solenoid valves, an IP54 enclosure, and a Rp 2.0M bill-of-materials constraint.

WikiScrolls App | *Flutter, Go, Neo4j, Docker, LLM*

- Co-developed an educational platform transforming Wikipedia articles into "TikTok-style" audiovisual content.
- Implemented frontend using Flutter and integrated LLM summarization with Text-to-Speech engines.

Akpro IME Web | *Astro, TypeScript, Figma*

- Designed UI/UX and developed the frontend for a centralized academic hub used by engineering students.
- Streamlined access to modules, schedules, and curriculum info, improving resource accessibility.

Draw Battle Game | *React, Express.js, PostgreSQL, Socket.io*

- Built a real-time multiplayer drawing and guessing game featuring live chat and synchronized canvas.
- Managed full-stack integration between React frontend and PostgreSQL backend via Socket.io.

Custom Linux Shell | *C Language, Linux CLI*

- Developed a functional CLI shell handling command execution, foreground/background process management, and IPC via pipes.

Network Infrastructure Design | *Cisco Packet Tracer, OSPF, EIGRP*

- Campus Network: Designed VLANs, subnetting, and OSPF routing for university-scale connectivity.
- WAN Architecture: Simulated Frame Relay and PPP topologies connecting HQ to branch offices.

AES-128 Hardware Accelerator | *VHDL, ModelSim, Quartus Prime*

- Designed a high-performance hardware accelerator for AES-128 encryption.
- Authored VHDL code for the core encryption process and validated integrity via testbenches.

Smart Drop-Box | *C++ (ESP32), FreeRTOS, Blynk IoT*

- Architected a secure package delivery system with OTP authentication and cloud monitoring.
- Orchestrated system concurrency using FreeRTOS and developed the mobile interface via Blynk.

Noir - Noise & Air Quality Monitor | *AVR Assembly, Proteus*

- Programmed sensor data processing in Assembly to control audio/visual alerts for environmental monitoring.

Parking Counter System | *Digital Logic, Proteus*

- Designed a logic circuit using counters and flip-flops to track parking availability in real-time.

TECHNICAL SKILLS

Languages: C/C++, C#, Java, Python, JavaScript, TypeScript, Go, Dart, VHDL, Assembly, LaTeX, Markdown

Web Development: React, Next.js, Astro, Express.js, FastAPI, Django, Flutter, HTML/CSS (Tailwind)

Database: PostgreSQL, MySQL, MongoDB, Redis, Neo4j

AI/ML & Vision: TensorFlow, PyTorch, OpenCV, Ultralytics YOLO

Networking & Cloud: Docker, Kubernetes, O-RAN, OpenAirInterface, FlexRIC, NVIDIA Aerial, Cisco Packet Tracer

Hardware & IoT: ESP32, Arduino, FreeRTOS, Proteus, ModelSim

Tools/Design: Git, Figma, Canva, Adobe, Trello, Jira

CERTIFICATIONS & AWARDS

English Proficiency: TOEFL ITP: 627 — EPT (UI): 643 (Index A)

Competitions: Technoskill 1.0 Web Dev (PIPTEK IME FTUI), Exertion Competitive Programming (EXERCISE FTUI)

Science Olympiads: Winner of Amaze with Science, National Science Olympiad (OSN) Math Participant